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COURSE PROJECT REPORT

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AI DUSTINO

THE AUTOMATIC PARKING GATE SYSTEM

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A project report submitted in partial fulfilment of the Requirements for the award of the degree of Bachelor of Computer Science

DATA STRUCTURE & ALGORITHMS

DEPARTMENT OF COMPUTER SCIENCE

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DECLARATION

We hereby declare that this project report is based on our original work except for citations and quotations which have been duly acknowledged. We also declare that it has not been previously and concurrently submitted for any other degree or award at Salim Habib University or other institution.

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1: INTRODUCTION

1.1 Background

In modern urban environments, parking management has become an important challenge due to the increasing number of vehicles. Manual parking gate systems require human intervention, which can lead to inefficiency, delays, and increased operational costs. Automation provides an effective solution to these problems by reducing human dependency and improving accuracy.

This project focuses on the design and implementation of an **Automatic Parking Gate System using Arduino and a Servo Motor**. The system automatically opens and closes a parking barrier when a vehicle is detected near the gate. The core objective of this project is to demonstrate the application of microcontroller-based automation using sensors, actuators, and control logic.

The project is developed as part of the **Computer Organization & Assembly Language (COAL)** course, emphasizing the relationship between high-level programming and low-level hardware execution. Concepts such as timing control, instruction execution, and hardware interfacing are reflected throughout the system design.

1.2 Problem Statement

Manual parking gate systems are inefficient and require constant human involvement, leading to delays and operational errors. Commercial automated solutions are costly and unsuitable for educational purposes. This project aims to design a low-cost automatic parking gate system using Arduino and a servo motor that operates autonomously while demonstrating microcontroller and COAL concepts.

1.3 Project Objectives

The primary objective of this project is to design and implement an **Automatic Parking Gate System** that operates autonomously using a microcontroller, a sensor, and a servo motor. The specific objectives of the project are as follows:

1. **To automate parking gate operation** by detecting the presence of a vehicle and controlling the opening and closing of the gate without human intervention.
2. **To design a low-cost and efficient system** using easily available components such as Arduino, ultrasonic sensors, and servo motors.
3. **To implement real-time decision-making logic** based on sensor input to ensure accurate and timely gate operation.
4. **To demonstrate microcontroller interfacing techniques**, including sensor input handling and actuator control.
5. **To apply concepts of Computer Organization & Assembly Language (COAL)** such as timing control, instruction execution, and hardware-level input/output operations in a practical system.
6. **To enhance understanding of embedded system design**, including system integration, control flow, and hardware–software interaction.

1.4 Scope

The scope of this project is limited to the development of a prototype automatic parking gate system using Arduino and a servo motor. It focuses on vehicle detection and automated gate control for educational purposes and does not include advanced commercial features such as access authentication or IoT integration

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2. PROJECT METHODOLOGY

2.1 System Architecture

The robot operates on a "Sense-Think-Act" paradigm. The system architecture is divided into three subsystems:

The robot operates on a **"Sense-Think-Act"** paradigm:

1. **Sensory Input:** The HC-SR04 Ultrasonic sensor continuously measures distance.
2. **Processing Unit:** The Arduino Uno (ATmega328P) processes distance data. If the distance is < 30cm, the logic triggers the actuator.
3. **Actuation:** The microcontroller sends a PWM signal to the Servo Motor to rotate the barrier arm.

2.2 Mechanical Design and Fabrication

The physical structure of the robot is a custom-engineered enclosure designed to house the electronics while maintaining a compact footprint.

2.3 Operational Workflow

1. System Initialization

- **Power-On:** Upon powering the system via the barrel jack or USB, the Arduino initializes the General Purpose Input/Output (GPIO) pins.
- **Setup:** The code attaches the servo motor to its control pin (typically Pin 9) and sets the barrier arm to the closed (0°) position.
- **Sensor Calibration:** The ultrasonic sensor pins (Trigger and Echo) are configured to their respective output and input modes.

2. Detection (Sense Phase)

- **Pulse Emission:** The Arduino triggers the HC-SR04 sensor to emit a high-frequency sonic pulse.
- **Echo Reception:** The sensor waits for the pulse to bounce off a vehicle and return, measuring the time taken for the echo to reach the receiver.
- **Distance Calculation:** The microcontroller converts this time duration into a distance measurement in centimeters.

3. Decision Logic (Think Phase)

Time Durations Used in the System

Accurate timing plays a critical role in the operation of the Automatic Parking Gate System. The ultrasonic sensor relies on precise microsecond-level delays to measure distance, while the servo motor operation depends on controlled timing to ensure safe gate movement. These timing mechanisms directly reflect core concepts of Computer Organization & Assembly Language (COAL), such as instruction execution time and hardware-level signal control.

3.1 Trigger Pulse Timing

The HC-SR04 ultrasonic sensor is activated by sending a short HIGH pulse to its trigger pin. This pulse causes the sensor to emit an ultrasonic sound wave that propagates through the air toward the approaching vehicle.

Signal State	**Duration**
LOW	2 microseconds
HIGH	10 microseconds

Purpose:
This trigger pulse initiates the transmission of the ultrasonic sound wave. The precise duration of the HIGH pulse is critical, as it ensures reliable sensor activation and accurate distance measurement.

3.2 Echo Time Measurement

After the ultrasonic pulse is transmitted, the sensor waits for the sound wave to reflect back from the detected object. The Arduino measures the duration for which the echo pin remains HIGH.

Code Statement Used:

```
duration = pulseIn(echoPin, HIGH);
```

- The variable duration stores the total time (in microseconds) taken by the ultrasonic wave to travel from the sensor to the object and return.
- This measured time represents the **round-trip travel time** of the ultrasonic signal.

The use of microsecond-level timing highlights the importance of precise instruction execution and hardware timing, which are fundamental concepts in Computer Organization and Assembly Language.

4. Actuation (Act Phase)

- **Gate Opening:** The Arduino sends a Pulse Width Modulation (PWM) signal to the servo motor, rotating the barrier arm from 0° to 120° (or 90°) to allow entry.
- **Passing Delay:** The system executes a timed delay (typically 1 to 2 seconds) to ensure the vehicle has sufficient time to pass under the barrier.
- **Gate Closing:** Once the delay is complete, the servo rotates back to its original 0° position to secure the entrance.

3. SOFTWARE REQUIREMENTS

The software architecture for the Automated Parking Gate System is designed to provide real-time responsiveness to vehicle detection. It bridges high-level control logic with low-level hardware timing, consistent with the objectives of the Computer Organization & Assembly Language (COAL) curriculum.

3.1 Integrated Development Environment (IDE)

- **Arduino IDE:** This is the primary environment used for writing, compiling, and uploading the source code to the ATmega328P microcontroller. It supports a specialized version of C++ and provides the necessary toolchain to convert code into machine-level instructions.
- **Fritzing (Optional for documentation):** This tool is utilized for creating clear, professional circuit diagrams and breadboard layouts, ensuring the physical wiring matches the software's pin definitions before assembly.

3.2 Programming Languages

- **C++ (Arduino):** The core logic is implemented in C++, utilizing the `Servo.h` library to manage Pulse Width Modulation (PWM) for precise motor positioning. It also handles the arithmetic required to convert sonic pulse duration into distance in centimeters.
- **Assembly Language (Contextual):** To align with COAL course concepts, the software relies on precise microsecond delays and register-level operations. Generating the 10µs trigger pulse for the ultrasonic sensor requires timing accuracy that mirrors Assembly-level instruction cycles to minimize latency.

3.3 Control Algorithm Logic

The software operates within a continuous `void loop()` structure, maintaining a "Sense-Think-Act" cycle to ensure no vehicle arrival is missed.

Pseudocode for Gate Operation:

1. **Read_Distance(Ultrasonic_Sensor):** Send a trigger pulse and measure the echo return time.
2. **Calculate(CM):** Convert time to distance.
3. **IF Distance < 30cm THEN:** (Vehicle detected at the gate)
    - **Rotate_Servo(120°):** Lift the barrier arm to the open position.
    - **Delay(2000ms):** Pause to allow the vehicle to pass through the gate.
    - **Rotate_Servo(0°):** Lower the barrier arm to the closed position.
4. **ELSE:** (No vehicle detected)
    - **Maintain_State:** Keep the servo at 0° and continue scanning.

4. HARDWARE REQUIREMENTS

This section details the specific components selected for the project, justifying their choice based on voltage requirements and torque needs.

4.1 Microcontroller

- **Arduino Uno R3 (with ATmega328P):** This serves as the "brain" of the gate system. It was selected for its reliability, 5V logic levels, and ease of interfacing with the servo and ultrasonic sensor. It handles the real-time I/O processing required to calculate distance and generate PWM signals.

4.2 Actuators

- **SG90 Servo Motor:** Unlike the DC motors used in mobile robots, a servo motor is selected for the gate to provide precise angular control (0° to 120°). It provides the necessary torque to lift the barrier arm smoothly and hold it in the "open" position without additional mechanical locking.
- **Barrier Arm (Chassis Attachment):** A lightweight custom-built or 3D-printed arm is coupled with the servo motor gear to act as the physical gate.

4.3 Sensors

- **HC-SR04 Ultrasonic Sensor:** This component uses sonar to determine the proximity of an approaching vehicle. It offers a detection range of 2cm to 400cm with 3mm accuracy, which is ideal for detecting a car within the 30cm threshold required to trigger the gate.

4.4 Power System

- **5V–9V Power Source:** The system can be powered via a battery pack or a regulated power adapter connected through the barrel jack interface.

- **Voltage Regulation:** The 5V logic for the ultrasonic sensor and the servo motor is supplied directly from the Arduino's regulated 5V pin, ensuring stable operation and protecting the sensitive logic circuits from voltage spikes.

4.5 Structural Components

- **Custom Case/Mount:** A stable base or 3D-printed housing is used to mount the ultrasonic sensor and the servo motor at the appropriate height for vehicle detection.
- **Jumper Wires and Breadboard:** These facilitate the electrical connections between the microcontroller, sensor, and actuator during the prototyping phase..

5. CONCLUSION

5.1 Summary of Achievements

This project successfully demonstrates the feasibility of creating a custom, autonomous parking barrier system using off-the-shelf electronic components. The system effectively integrates ultrasonic sensing with servo motor actuation to manage vehicle entry in a compact, portable unit. The prototype proved ideal for simulating real-world parking scenarios, and the ultrasonic obstacle avoidance logic functions reliably to trigger the gate only when a vehicle is within the specified safety threshold.

5.2 Challenges Encountered

- **Power Distribution:** Managing the electrical load between the servo motor's current draw and the 5V logic circuit of the Arduino was critical to prevent voltage drops or noise interference during gate movement.
- **Signal Calibration:** Establishing the correct threshold for the ultrasonic sensor was challenging due to potential "ghost" echoes in narrow environments, requiring precise software filtering.
- **Timing Accuracy:** Aligning the gate's "open" duration with the average time required for a vehicle to pass necessitated multiple iterations of the delay logic to ensure safety.

5.3 Future Recommendations

To enhance the system into a commercial-grade smart parking solution, future iterations could include:

- **LiDAR Mapping:** Implementing advanced scanning for intelligent vehicle detection and area monitoring rather than simple proximity logic.
- **RFID/Infrared Integration:** Adding infrared receivers or RFID scanners to allow the gate to autonomously recognize authorized vehicles for secure entry.

- **Mobile App & IoT Integration:** Using an ESP32 Wi-Fi module to allow facility managers to monitor parking occupancy or manually override the gate via a smartphone.

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